

Twin Pregnancy 01. Diagnosis, Chorionicity, and Amnionicity

Companion reading handout for simultaneous listening.

Audio course on twin pregnancy. Episode one. The twin diagnosis is not enough.

Source base

This episode is based on Gabbe chapter 32 on multiple gestations, the Israeli position paper on twin pregnancy follow-up, and the Israeli ultrasound position paper.

Opening frame

Let us begin with the central idea.

There is no such thing as a complete diagnosis of "twins." That word tells you the number of fetuses, but it does not tell you the biology of the pregnancy. The clinically meaningful diagnosis is dichorionic diamniotic twins, monochorionic diamniotic twins, or monochorionic monoamniotic twins.

That distinction is not clerical. It predicts the complication profile, the follow-up schedule, the counseling, and the timing and route of delivery.

Key point

First, separate two words that are often confused.

Zygoty means genetic origin. Dizygotic twins come from two eggs and two sperm. Monozygotic twins come from one fertilized egg that divides.

Chorionicity means placental architecture. It answers a different question. Are there two chorions and usually two placentas, or is there one shared chorion and therefore one shared placental territory?

For management, chorionicity matters more than zygoty. Monozygoty is interesting. Chorionicity is dangerous.

Dizygotic twins are almost always dichorionic diamniotic, because each embryo builds its own chorion and its own amnion. Monozygotic twins depend on timing. If the embryo splits very early, before about 72 hours, the result is dichorionic diamniotic. If it splits from day 4 to day 7, the result is monochorionic diamniotic. If it splits from day 8 to day 12, the result is monochorionic monoamniotic. Very late splitting can produce conjoined twins.

This is the first pathophysiologic anchor. The later the split, the more shared the environment.

A dichorionic pregnancy is essentially two pregnancies sharing the uterus. A monochorionic pregnancy is different. It is two fetuses sharing one placenta, and usually sharing vascular communications inside that placenta. That is why twin-to-twin transfusion syndrome, or TTTS, twin anemia polycythemia sequence, or TAPS, and selective fetal growth restriction, or sFGR, belong to the monochorionic world.

Now, how do we diagnose this correctly?

Early ultrasound is the key. In the first trimester, you can often count gestational sacs, yolk sacs, embryos, and membranes before anatomy becomes crowded and before membranes become harder to see.

The ultrasound report in a multiple pregnancy should not merely say "viable twins." It should document the number of gestational sacs, their location, and the chorionicity and amnionicity. The Israeli ultrasound position paper is explicit about this.

Documentation matters because the first-trimester image may be the most reliable evidence available later.

In early pregnancy, two separate sacs with a thick dividing membrane support dichorionic diamniotic twins. A single gestational sac with two fetuses and a thin dividing membrane supports monochorionic diamniotic twins. No dividing membrane, with both fetuses in one amniotic space, supports monochorionic monoamniotic twins, although this diagnosis may not be secure very early.

At 11 to 14 weeks, the base of the intertwin membrane becomes very useful. A lambda sign, also called a twin-peak sign, supports dichorionicity. It represents chorionic tissue extending between the two amniotic sacs. In monochorionic twins, there is no chorion between the sacs, so the membrane meets the placenta as a thin T-shaped junction.

Do not reduce this to memorizing lambda versus T. Understand the anatomy. If chorionic tissue enters the membrane base, the pregnancy is dichorionic. If the two amnions meet one shared chorion without intervening chorion, the pregnancy is monochorionic.

After the early second trimester, determination becomes less reliable. The placentas may fuse. A single placental mass may look monochorionic even if the pregnancy is dichorionic. A thin membrane may be missed. Same-sex fetuses remove one useful clue. Different fetal sex strongly supports dizygotic, and therefore dichorionic, twins. But same sex does not prove monochorionicity.

This is why the Israeli twin follow-up paper gives a conservative rule. If a twin pregnancy is diagnosed in the second trimester or later, and chorionicity cannot be determined confidently, manage it as monochorionic diamniotic.

That rule is not defensive medicine. It is biologically rational. The harm of under-calling monochorionicity is missing TTTS, TAPS, or sFGR. The harm of over-calling monochorionicity is more intensive surveillance. For an uncertain same-sex twin pregnancy with a single placental mass, safety favors the monochorionic pathway.

Now let us pause and organize the categories

Now let us pause and organize the categories.

DCDA means dichorionic diamniotic. Two chorions, two amniotic sacs. This group still has increased risk of preterm birth, growth problems, preeclampsia, cesarean delivery, and neonatal morbidity compared with singleton pregnancy. But it does not have placental vascular-anastomosis diseases.

MCDA means monochorionic diamniotic. One chorion, two amniotic sacs. These twins can move in separate sacs, but they share one placenta. The membrane protects them from cord entanglement, but not from shared vascular circulation.

MCMA means monochorionic monoamniotic. One chorion, one amniotic sac. These fetuses share the placenta and the amniotic space. That adds cord entanglement and cord accident risk to the monochorionic vascular risks.

The resident-level way to remember this is simple. Chorionicity predicts placental disease. Amnionicity predicts cord-space disease.

There is one more distinction worth making early: fetal labels.

Twin A and twin B are not casual names. They are clinical identifiers. Later procedures, growth curves, Dopplers, amniocentesis, selective reduction, and delivery decisions all depend on correct labeling. The fetus closer to the cervix is usually called twin A, but the label should be anchored with placental location, sac position, fetal sex if different, cord insertion, and any structural marker. Poor labeling is not a documentation problem. It can become a wrong-fetus problem.

Now final synthesis

Now final synthesis.

First, the word twins is incomplete. The real diagnosis is DCDA, MCDA, or MCMA.

Second, zygosity is genetic origin, but chorionicity is the management variable. Monochorionicity is the risk amplifier.

Third, first-trimester ultrasound is the best time to assign chorionicity and amnionicity. Later uncertainty should usually be managed as monochorionic diamniotic.

Fourth, chorionicity predicts placental vascular complications, and amnionicity predicts cord-space complications.

If you keep that framework, the rest of twin pregnancy becomes much easier. Surveillance, complications, and delivery timing are not separate lists. They are consequences of placental architecture.

End of episode one.